Yuxin Ye

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INDUSTRY EXPERIENCE

Alibaba Group, Tongyi Lab (Qwen) - Algorithm Engineer, RL Infrastructure

Mar 2026 - Present

- Parallelize large-scale ASR and audio-encoder pretraining for Qwen omni models.
- Build GRPO / PPO post-training infrastructure for Qwen MoE models on verl, connecting Megatron training and vLLM inference through a separated trainer-rollout architecture and NCCL / NIXL weight synchronization.
- Engineer reward systems combining rule-based verifiers and LLM-as-judge evaluation for accuracy, instruction-following, and helpfulness.
- Build two-stage vLLM-omni rollout with encoder/decoder disaggregation and shared-memory transfer of hidden states, supporting trainable-encoder GRPO.
- Develop agent tool-execution sandbox integration on Alibaba Cloud ACS / Kubernetes, including E2B SDK adapters, reproducible tool replay, artifact collection, and environment cleanup. Validated 208 command replays across 16 fresh CPU environments.
- Build and debug long-horizon agent rollout and evaluation harnesses for APEX and SWE-bench, including tool-call protocols, context / token budgeting, final-answer handling, grading coverage, and trajectory observability.

Cerebras Systems - Member of Technical Staff

Jan 2025 - Mar 2026

- Built RL post-training on wafer-scale hardware (CSX / WSE): wrote the trainer API against its instruction set, memory model, and compiler stack, and stood up a verl-based RL stack for training-inference co-design on non-GPU silicon.
- Deployed and debugged datacenter-scale Cerebras clusters for reliable large-model training and inference.

Lamini AI - High Performance Computing Engineer

May 2024 - Jan 2025

- Co-authored *Banishing LLM Hallucinations Requires Rethinking Generalization* (Lamini Memory Tuning) with Dr. Gregory Diamos.
- Enabled RoCE RDMA over AMD MI300 / MI250 with Mellanox / Broadcom drivers and Kubernetes / Slurm; implemented DP and FSDP with mpi4py over a 100 Gbps RoCE network; deployed vLLM-compatible serving across NVIDIA and AMD machines.

Amazon Robotics - Software Development Engineer

Jul 2022 - Mar 2024

- Developed features for an event-driven, multi-agent simulator of automated sortation floors, including multi-robot drive-path planning, staging-cell distribution, and resource-based allocation.

PUBLICATION

Johnny Li, Saksham Consul, Eda Zhou, James Wong, Naila Farooqui, Yuxin Ye, Nithyashree Manohar, Zhuxiaona Wei, Tian Wu, Ben Echols, Sharon Zhou, Gregory Diamos. *Banishing LLM Hallucinations Requires Rethinking Generalization.* arXiv:2406.17642, 2024.

EDUCATION

Harvard University, GSAS - M.S., Computational Science and Engineering · 2020 - 2022 · GPA 3.87

University of Science and Technology of China - B.S., Physics, School of the Gifted Young · 2016 - 2020 · GPA 3.7

SELECTED PROJECTS

Draft Models as Test-Time-Trained Search Policies

Ongoing research

Investigate a small draft model as both a speculative-decoding proposer verified by a larger target and a rollout policy for MCTS over chain-of-thought. Study per-problem adaptation using search outcomes, with evaluation designed to compare accuracy at matched inference compute. This is an ongoing research direction.

Gomoku Solver - Monte Carlo Tree Search

2021 - 2022 · MIT 6.877 Principles of Autonomy and Decision Making

Implemented UCT MCTS for 9x9 Gomoku with a neural network as the default rollout policy.

Reinforcement Learning of Soft Fish-like Swimmers (C++)

2021 - 2022 · Research Assistant, Prof. Petros Koumoutsakos, Harvard SEAS

Coupled fluid-soft-body interaction using the inverse-map technique into a direct numerical flow simulator built on the Cubism library with Adaptive Mesh Refinement. Parametrized swimmer gait with neural spline knots and trained swimming behavior via reinforcement learning over muscle actuation. Code

Wasserstein Learning of Generative Models

Apr - May 2021 · MIT 6.838 Shape Analysis

Built WGAN-GP from scratch, validated its tractable loss and a Monge-map construction, and benchmarked against W2GAN on MNIST and CIFAR-10.

Realtime Snow Simulation and Rendering (C++)

Nov - Dec 2020 · MIT 6.837 Computer Graphics · Course project contest runner-up

Simulated snow with the moving-least-squares material point method (MLS-MPM) and rendered in real time.

Auto-differentiation Package (C++)

Oct - Dec 2020

Implemented automatic differentiation in forward and reverse modes, including expression-graph construction and derivative transforms. Code

TEACHING

Teaching Assistant - CS 205: Computing Foundations for Computational Science (HPC / HTC), Harvard GSAS · Spring 2022

RESEARCH INTERESTS

Reinforcement pretraining and verifiable rewards; test-time compute and search-guided self-improvement; predictive representations and encoder compression; inverse RL for alignment; discrete flows and non-autoregressive generation; IO and data movement in architecture design.

SKILLS

Methods: RL post-training (GRPO / PPO / DPO), inverse RL, MCTS and search, speculative decoding, test-time training, flow matching, reward modeling.

Training and inference: PyTorch, JAX, Megatron-LM, DeepSpeed, FSDP, verl, vLLM.

GPU and communication: CUDA, ROCm, Triton; NCCL, NIXL, RDMA (RoCE), MPI.

Languages and infrastructure: C/C++, Python, Go, Julia; Linux, Kubernetes, Slurm, Git.